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Prediction and optimization of structural performance of prefabricated bridges based on physical information neural network (PINN) and BIM

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Abstract
This study presents an innovative performance prediction and optimization method for prefabricated bridge structures, which integrates the strengths of Physical Information Neural Network (PINN) and Building Information Modeling (BIM). By combining the computational power of PINN for solving structural mechanics problems with BIM's data management and modeling capabilities in construction engineering, a multi-layer, multi-dimensional intelligent framework for prediction and optimization is developed. BIM models provide crucial data, including geometric details, material properties, and construction process information, which are used to create a digital twin of the bridge. Leveraging PINN's robust fitting ability, physical laws such as stress, strain, and equilibrium equations are incorporated into the neural network to predict the bridge's performance under various operational conditions. The integration of physical constraints during PINN training enhances both the accuracy and reliability of the predictions. Subsequently, optimization algorithms refine the bridge design based on these predictions, ensuring improvements in performance while meeting key criteria such as safety, cost-effectiveness, and environmental impact. This method offers a scientifically rigorous and highly accurate approach to supporting the design and construction of prefabricated bridges, advancing intelligent construction and sustainable building practices. Experimental results demonstrate that the integrated PINN-BIM model outperforms traditional methods, exhibiting superior accuracy, generalization capability, and offering a novel solution for efficient bridge engineering construction and performance optimization.

Keywords Physical information neural network (PINN) · Building information modeling (BIM) · Prefabricated bridges · Performance prediction · Structural optimization · Intelligent construction

1 Introduction

With the continuous advancement of global infrastructure construction, prefabricated bridges have received widespread attention as an efficient, environmentally friendly, and sustainable construction method. Prefabricated bridges not only greatly shorten the construction period but also reduce the impact on the environment during the construction process by prefabricating various bridge components in the factory and transporting them to the site for assembly. However, despite the many advantages of prefabricated bridges, the prediction and optimization of their

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structural performance still face many challenges. Traditional structural design methods often rely on empirical formulas, finite element analysis, and other methods. Although they can provide some evaluation of structural performance, they still have certain limitations when facing complex mechanical behavior, uncertainty during construction, and the influence of different environmental conditions. Especially when considering the nonlinear behavior of materials, construction sequence, and coupling of multiple physical fields, traditional methods often struggle to achieve comprehensive and accurate performance prediction and optimization [1].

In recent years, Physics-Informed Neural Networks (PINNs) have gradually become a powerful tool for solving complex engineering problems as an emerging machine learning technology. PINNs combine physical laws with neural networks to effectively model systems and make efficient predictions and optimizations based on physical equations, especially in situations where data is scarce. The emergence of this technology provides new ideas for the performance analysis of bridge structures [2]. On the other hand, Building Information Modeling (BIM), as an integrated digital modeling tool, has been widely used in the construction industry, especially in bridge design and construction management. BIM can provide accurate geometric modeling, construction planning, and material information, thereby helping designers, engineers, and construction personnel to collaborate and make better decisions. However, BIM itself still has certain limitations in the in-depth analysis of structural performance and multi-physics coupling simulation [3].A significant advancement in addressing these limitations is the integration of live data measurement systems directly installed on structures, which enables the continuous monitoring of real-time performance data. By incorporating these sensor-based measurements into BIM, it becomes possible to create more trustworthy and dynamic models that reflect the actual behavior of the structures under varying conditions [4]. This approach not only enhances the accuracy of BIM models but also strengthens their predictive capabilities. The integration of real-time monitoring data is gaining increasing global recognition as an essential practice for enhancing the reliability and functionality of BIM, particularly in the context of structural performance analysis. This practice is gaining traction due to its potential to bridge the gap between theoretical modeling and actual physical performance, thus providing more reliable insights for maintenance, safety, and optimization [5].Physics-Informed Neural Networks.

Therefore, combining PINN with BIM technology to form a new intelligent design and optimization method can make up for the shortcomings of traditional methods. By combining the precise structural data provided by BIM with the physical constraints of PINN, mechanical behavior, dynamic changes during construction, and external environmental influences can be fully considered in performance prediction and optimization, thereby achieving more accurate and comprehensive prediction and optimization of bridge structural performance. This can not only enhance the safety and reliability of bridge design, but also provide more scientific and systematic decision support for the construction of prefabricated bridges. Therefore, this article aims to explore the theoretical framework and application path of combining PINN and BIM technology, provide new technical solutions for predicting and optimizing the structural performance of prefabricated bridges, and provide theoretical support and methodological guidance for practical problems in engineering practice.

In this study, the term "performance" refers to a comprehensive set of mechanical and operational characteristics of bridge structures. Specifically, we focus on several key aspects:

- ✣ Stress State: The distribution and magnitude of stresses within the bridge components under various loading conditions.
- ✣ Deformations and Displacements: The geometric changes and movements of the bridge structure, including deflections and rotations, under different operational scenarios.
- ✣ Operability: The ability of the bridge to function effectively under normal usage conditions, including traffic loads and environmental influences.
- ✣ Extreme Events: The structural behavior and resilience of the bridge during extreme conditions such as high winds, earthquakes, or other natural disasters.

By addressing these aspects, our method aims to provide a holistic view of the bridge's performance, ensuring that the design and optimization processes consider all critical factors that affect the bridge's safety, durability, and efficiency.

2 Related work

2.1 Development and application of physics-informed neural networks (PINN)

Physical Information Neural Network (PINN) is an innovative method that integrates physical laws and deep learning techniques [6]. It embeds physical equations directly into the training process of neural networks (NNs), enabling the model to not only learn patterns from data but also automatically satisfy physical constraints during the solving process. This approach allows for a more accurate and complex representation of practical engineering problems. In this context, physical laws refer to the fundamental equations that describe physical phenomena, and deep learning refers to advanced algorithms that simulate the human brain's neural structure for pattern recognition and problem-solving. The architecture of the Physical Information Neural Network (PINN) is shown in Fig. 1.

In Fig. 1, "PDE" denotes the partial differential equation, "BC" denotes the boundary condition, and "IC" denotes the initial condition.

Unlike traditional deep learning methods, PINN does not rely on a large amount of annotated data, but can ensure the physical consistency of the model output by incorporating known physical laws (such as fluid mechanics equations, structural mechanics equations, etc.) into the loss function of the network. This method is particularly suitable for solving problems that are difficult to efficiently solve through traditional numerical methods, especially when dealing with complex boundary conditions, nonlinear phenomena, or large-scale data. PINN exhibits unique advantages. In recent years, PINN has made significant progress in multiple engineering fields, especially in fluid mechanics, structural mechanics, and other fields, and has begun to be widely applied. In fluid mechanics, PINN is used to solve complex fluid dynamics problems such as Navier Stokes equations, achieving efficient flow simulation and prediction. In the field of structural mechanics, PINN also performs well, especially in bridge engineering, and has gradually been applied in structural health monitoring, stress analysis, material fatigue prediction, and other aspects. For example, researchers use PINN to monitor the stress state of bridge structures in real time, which can effectively predict the deformation behavior of bridges under different load conditions, identify possible structural failures in advance, and provide scientific basis for bridge maintenance and management. In addition, PINN is also applied in the bridge design phase to help engineers optimize structural design, reduce material waste, and improve the safety and economy of the design [7].

In the future, with the improvement of computing power and the development of artificial intelligence technology, the application prospects of PINN in the engineering field will become increasingly broad. In the future, PINN will not only provide more accurate performance predictions, but also play an important role in optimizing design, construction management, and other aspects. Especially when combined with advanced technologies such as Building Information Modeling (BIM), it will further promote the development of intelligent construction and improve the overall performance and efficiency of bridge engineering.

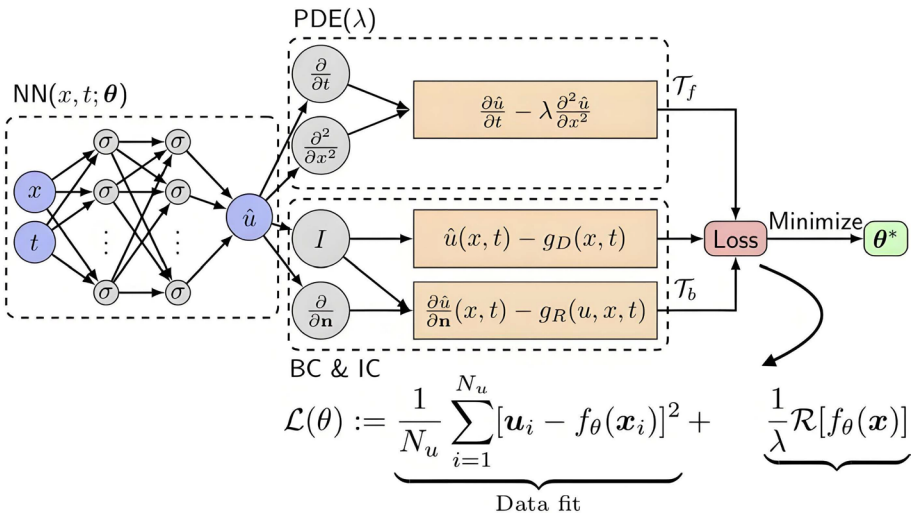


Fig. 1 Physical Information Neural Network (PINN) architecture

2.2 Building information modeling (BIM) and bridge structure design

Building Information Modeling (BIM) is a technology-driven approach that integrates detailed data from the design, construction, and operation stages of buildings through digital and 3D modeling technologies. This integration greatly enhances the efficiency and accuracy of bridge structure design. In the traditional bridge design process, designers often rely on manually drawn drawings and their experience to carry out scheme designs. However, this method is prone to information asymmetry or deviation during construction and maintenance, leading to wasted project costs and time. The introduction of BIM facilitates collaborative management and real-time updates of all data, materials, construction methods, and post-operation processes related to bridges on a unified platform. This ensures the accuracy and traceability of the design, construction, and operation processes.

In this context:

BIM stands for Building Information Modeling, a digital representation of physical and functional characteristics of a building or infrastructure.

3D Modeling Technologies refer to software tools and techniques that create three-dimensional digital models of structures.

Design, Construction, and Operation stages represent the entire lifecycle of a building or infrastructure project, from conceptualization to completion and ongoing maintenance.

Information Asymmetry refers to a situation where some stakeholders possess more or different information than others, often leading to errors or inefficiencies.

Collaborative Management involves multiple stakeholders working together, often in real-time, to manage the project's data and progress.

BIM Bridge 3D Modeling is shown in Fig. 2.

In bridge engineering, BIM can not only optimize structural design, but also simulate the construction process of different design schemes, predict possible risks and problems [8]. For example, in the design phase, BIM can help engineers create precise 3D models, evaluate the safety, stability, and economy of different bridge design schemes, and determine the best design solution in the shortest possible time [9]. During the construction process, BIM can also serve as a guide for construction personnel, providing accurate construction paths, schedule planning, and material management information, effectively reducing errors and rework, and ensuring the controllability of construction quality and progress [10]. BIM still plays an important role in the operation stage of bridges, by monitoring the structural health status of bridges in real time, warning possible faults in a timely manner, guiding bridge maintenance and repair work, thereby extending the service life of bridges and reducing maintenance costs [11]. Especially in the design of prefabricated bridges, BIM plays a crucial role [12]. Prefabricated bridges rely on prefabricated components and modular design, which can significantly improve construction efficiency and reduce on-site construction uncertainty [13]. Through BIM technology, designers can accurately simulate the installation process of prefabricated components, optimize the size and combination of components, and ensure the accuracy of assembly and smooth construction [14]. In addition, BIM can effectively manage the material requirements and inventory of prefabricated bridges, avoid resource waste, and



Fig. 2 BIM Bridge 3D Modeling

track the status of each component in real time during the actual construction process, thereby improving the safety and controllability of construction [15].

With the development of Physics-Informed Neural Networks (PINN), the combination of BIM and PINN in bridge structure design and optimization will bring revolutionary progress. PINN can ensure further improvement in design accuracy and optimization solutions based on BIM design by introducing physical laws and equations. For example, when predicting the performance of bridge structures, PINN can directly embed physical equations such as fluid mechanics and structural mechanics to ensure that the calculation results conform to actual physical laws. This fusion method enables real-time evaluation of the performance of bridges under different loads and environmental conditions during the design phase, providing more accurate predictions and optimization suggestions, thereby achieving more efficient and safe management and decision-making in the design, construction, and operation of bridges.

2.3 Performance prediction and optimization methods for prefabricated bridge structures

The performance prediction and optimization methods of prefabricated bridges traditionally rely on finite element analysis (FEA) and optimization algorithms. These methods can predict the stress behavior and deformation of bridges to a certain extent through discretization of bridge structures, simplification of boundary conditions, and assumptions, but there are still some limitations [16].

Firstly, finite element analysis requires a large amount of computational resources when dealing with complex geometric shapes and nonlinear material behavior, especially in bridge structure optimization, where the computational workload grows exponentially, resulting in significant computation time and resource consumption [17]. Secondly, traditional methods often rely on simplifying assumptions when dealing with complex boundary conditions or multiple influencing factors, which may lead to bias and low accuracy in the results, especially in high-level design stages [18]. In addition, the finite element method also faces certain challenges in multi-objective optimization or real-time scheduling, and cannot quickly respond to changes in engineering requirements [19]. In recent years, data-driven artificial intelligence technologies, especially deep learning and neural network (NN) methods, have been widely applied in the field of bridge structure analysis and optimization [20]. These technologies utilize a large amount of experimental data, on-site monitoring data, or simulation data to more accurately describe the structural performance of bridges and provide intelligent decision support for bridge design and optimization [21]. Neural networks can automatically extract potential patterns and patterns from large amounts of data by learning the mechanical properties and dynamic behavior of bridge structures, thus compensating for the shortcomings of traditional analysis methods [22]. On this basis, the combination of Physical Information Neural Network (PINN) and Building Information Modeling (BIM) has opened a new chapter in predicting and optimizing the performance of prefabricated bridge structures [23]. PINN can improve the accuracy and reliability of models while maintaining computational efficiency by embedding physical laws such as mechanics equations, heat conduction equations, etc. into neural networks [24]. It can predict structural performance in real-time through data-driven methods, combined with the actual physical characteristics of the bridge [25]. For example, PINN can quickly predict the stress and strain distribution of bridges under different working conditions and possible structural problems by inputting their geometric parameters, material properties, and load conditions, providing a basis for subsequent design optimization. BIM, as the core platform for prefabricated bridge design and construction, provides rich data support for the application of PINN. Through BIM models, engineers can accurately obtain geometric and material information of various parts of the bridge, as well as the interrelationships between components. These data can be directly input into the PINN model to achieve more accurate performance prediction. In addition, the BIM platform can integrate dynamic data such as bridge construction progress, material usage, and environmental changes, providing PINN with real-time updated data sources to further improve the accuracy of prediction and optimization.

In short, combining BIM and PINN methods can not only provide more accurate performance predictions during the design phase, but also monitor the structural status of bridges in real-time during the construction phase, and optimize the service life and safety of bridges through prediction and maintenance plans during the operation phase. Performance in this context includes factors such as stress states, deformation, and displacement, and these factors are evaluated across different stages: during the construction phase, where the structural integrity and adaptability to construction processes are critical; during the operability phase, focusing on the bridge's performance under normal usage and load conditions; and in the case of extreme events, where the behavior of the structure under unusual or disaster conditions is also assessed. This fusion method provides intelligent management support for the design, construction, and maintenance of prefabricated bridges throughout their entire lifecycle, making bridge design and construction more efficient, economical, and safe.

2.4 Main research content and innovative points of this article

This article proposes a method for predicting and optimizing the performance of prefabricated bridge structures by combining Physics-Informed Neural Networks (PINN) with Building Information Modeling (BIM), aiming to enhance decision support capabilities during bridge design, construction, and maintenance phases. By integrating the high-precision geometric, material, and construction information provided by BIM, as well as the advantages of PINN in handling physical laws and data-driven models, this paper proposes a new intelligent framework that can improve the performance prediction accuracy and optimization effect of prefabricated bridges while ensuring computational efficiency. This method not only considers the static and dynamic behavior of the structure, but also dynamically adjusts prediction and optimization schemes to cope with complex working conditions and environmental changes, thereby achieving full lifecycle management of the bridge.

The innovation of this article is mainly reflected in the following three aspects:

1) Designed a framework that integrates PINN and BIM, fully utilizing the precise data provided by BIM models to guide the construction and training of PINN models, overcoming the shortcomings of traditional methods in geometric and material property accuracy;
2) A deep learning optimization method based on physical constraints is proposed, which embeds physical laws into the deep learning network, improves the accuracy of structural performance prediction, and avoids the overfitting problem that may occur in data-driven models;
3) The effectiveness of this method in optimizing the performance of bridge structures has been verified through experiments, demonstrating its feasibility and superiority in practical applications, providing new ideas and technical support for the intelligent design and construction of bridge engineering.

3 Bridge structural performance prediction method integrating PINN and BIM

3.1 Design of PINN and BIM Fusion framework

In order to achieve accurate prediction and optimization of prefabricated bridge structures, this paper designs a framework that integrates Physical Information Neural Network (PINN) and Building Information Modeling (BIM). This framework consists of two core steps: data integration and model training. It can effectively combine the high-precision geometric information and material properties provided by BIM, while utilizing the advantages of PINN in processing physical laws and data-driven models to improve the accuracy and efficiency of structural performance prediction.

In this context:

Geometric Information represents the precise measurements and shapes of the bridge structure, typically provided in 3D models.

Material Properties include the characteristics of the materials used in the construction of the bridge, such as strength, elasticity, and durability.

Physical Laws are fundamental equations that describe the natural behavior of materials and structures under different conditions.

Data-driven Models refer to machine learning or artificial intelligence algorithms that extract patterns from data to predict outcomes without explicit physical formulas.

Model Training involves the process of using data to improve the performance of a predictive model by adjusting its parameters.

Data Integration is the process of combining different types of data (e.g., geometric, material properties) into a unified format that can be processed by the model.

The PINN and BIM Fusion Framework is shown in Fig. 3.

Firstly, BIM provides basic data such as detailed geometric shapes, material properties, and loading conditions of bridges [26–28]. These information include but are not limited to node coordinates, component dimensions, elastic modulus and density of materials, prestressing parameters, etc. Assuming that the bridge structure can be represented as a set of discrete nodes and a collection of components connecting these nodes, the geometric shape can

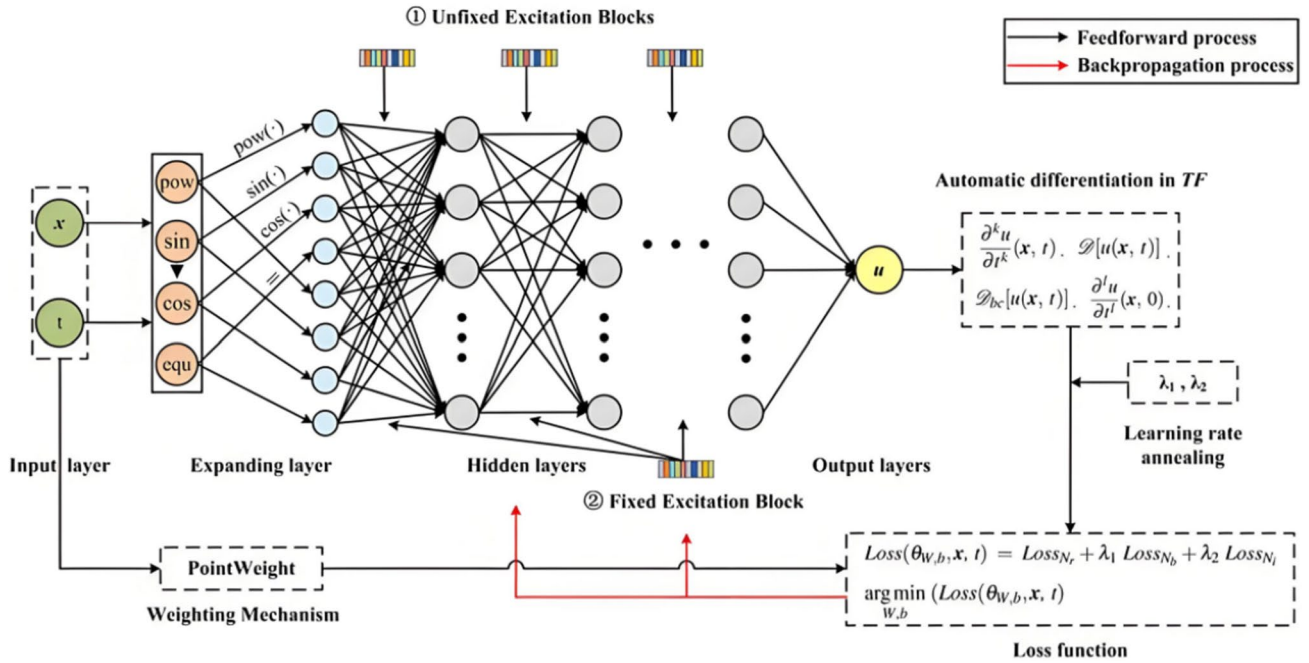


Fig. 3 PINN and BIM Fusion Framework

be represented as a three-dimensional spatial model through BIM data. We set the geometric information of the structure to $\left(X = x_1, x_2, \ldots, x_N\right)$, where $\left(x_i\right)$ represents the (i) position of the node.

Next, (ρ) the material properties provided in BIM, such as elastic modulus(E), Poisson's ratio, (ν) and density, are used to define the physical properties of the material, which will be fed into the PINN model as inputs. The constitutive relationship of materials usually follows classical elastic theory, assuming that under the assumption of small deformation, the stress–strain relationship of the structure can be described by linear elastic theory:

$$\sigma = C\epsilon \tag{1}$$

Among them, (σ) represents the (C) stress tensor, (ϵ) is the elastic matrix of the material, and is the strain tensor. The elastic matrix of a material is (C) usually (ν) calculated by its elastic modulus (E) and Poisson's ratio using the following formula:

$$C = \frac{E}{(1+\nu)(1-2\nu)} \begin{bmatrix} 1-\nu & \nu & 0 \\ nu & 1-\nu & 00\ 0\ \frac{1-\nu}{2} \end{bmatrix} \tag{2}$$

Once these data are integrated, they will be $(PINN)$ combined with Physics-Informed Neural Networks and begin to enter the model training phase. The core idea of PINN is to combine physical equations with neural networks, utilizing the powerful fitting ability of neural networks and the constraints of physical equations to guide network optimization during the training process. Specifically, PINN maintains physical consistency in prediction results by introducing equilibrium equations and material constitutive relationships from structural mechanics.

The static equilibrium equation of a structure can be expressed as:

$$Ku = F \tag{3}$$

Among them, (K) is the stiffness matrix of the structure, (u) is the displacement vector of the nodes, and (F) is the external load vector. PINN not only minimizes the loss function during training and compares the error between the network output and actual observed data, but also requires the neural network's prediction results to satisfy the above physical equations [29, 30]. In this way, PINN can learn structural responses that conform to physical laws by adjusting the weights of the network.

In addition, during the model training process, the network parameters are optimized using the following constraints to ensure that the predicted displacement and stress distributions meet the boundary and loading conditions. Finally, in the optimization phase, the output of the PINN model is backpropagated using gradient descent to minimize the total loss function, which includes prediction error, physical constraint error, and regularization term. The optimization objective can be written as:

$$L = L_{\text{data}} + L_{\text{physics}} \tag{4}$$

Among them, (L_{data}) represents the data-driven error term, (L_{physics}) represents the physical constraint error term, (λ) is the regularization coefficient used to balance the weights of the two.

Through the integration of PINN and BIM framework, this article can accurately predict the performance of prefabricated bridges under different working conditions and provide effective technical support for subsequent structural optimization.

3.2 Construction of PINN Model for prefabricated bridge structures

It is crucial to construct a model based on Physical Information Neural Network (PINN) in the process of predicting and optimizing the performance of prefabricated bridge structures [31, 32]. In order to improve the accuracy and robustness of the model, this paper embeds various factors such as material properties, boundary conditions, and damage deformation of bridge structures into the PINN model, thereby achieving accurate prediction of bridge performance under different working conditions. Meanwhile, an adaptive method is adopted to model the damage and deformation behavior of bridges during service, in order to adapt to the unique construction and working characteristics of prefabricated bridges [33, 34].

In the specific construction process of the model, the material characteristics of the prefabricated bridge need to be integrated into the PINN model first. The materials of bridges can generally be divided into reinforced concrete, prestressed concrete, and other composite materials, and the constitutive relationship of these materials is crucial for the overall response of the structure [35]. Considering the response of bridges in different working states, the PINN model should simultaneously handle the linear and nonlinear material properties of bridges. Taking linear elastic materials as an example, the stress–strain relationship of the structure can be expressed as:

$$\sigma = C\epsilon \tag{5}$$

Among them, (σ) is the (C) stress tensor, (ϵ) is the elastic matrix of the material, and is the strain tensor. For nonlinear materials [36], such as bridges with cracks or plastic regions, a nonlinear constitutive model, such as the Ramberg Osgood model, can be introduced to describe the stress–strain relationship of the material under large deformations:

$$\sigma = E\epsilon(1 + \alpha\epsilon^{n}) \tag{6}$$

Among them, (α) and (n) are nonlinear parameters that reflect the hardening behavior of the material.

Furthermore, considering the relative slippage and damage between nodes and components of prefabricated bridges [37, 38], the model must be able to take into account the effects of these behaviors. Therefore, the damage evolution model is introduced into the PINN framework. The damage variable is (D) described by a damage evolution equation, such as modeling through the following constitutive relationship:

$$\frac{dD}{dt} = f(\sigma, D) \tag{7}$$

Among them, the $(f(\sigma, D))$ driving function representing the evolution of damage can be derived from experimental data or theoretical analysis. The damage behavior is not only related to the properties of the material itself, but also closely related to the loading history of the structure and environmental factors [39]. For prefabricated bridges, damage in the connection area is usually manifested as slip and plastic deformation at the joints. Therefore, the model needs to introduce an additional joint damage variable (D_j) and describe it through a damage index:

$$D_j = \frac{\sigma_{max}}{\sigma_{yield}} \tag{8}$$

Among them, (σ_{yield}) (σ_{max}) is the maximum stress measured during the service of the bridge, which is the yield strength of the material. In this way, the model can describe the damage and deformation caused by material fatigue, temperature changes, and load fluctuations during construction and operation.

In order to further improve the adaptability of the model to prefabricated bridges under complex working conditions, we introduced adaptive methods to handle nonlinear problems and complex boundary conditions in the model. The adaptive method automatically adjusts the network weights based on the damage state and deformation situation, enabling PINN to optimize the structural response during training based on new observation data and boundary conditions [40, 41]. Assuming the displacement field of the $(u(x,t))$ structure is, after considering the adaptive strategy, the motion equation of the structure can be expressed as:

$$M\partial t2\partial 2u + K(u)u = F(u,t) + D(u)(Transition\ Equation) \tag{9}$$

Among them, $(K(u))$ is the structural stiffness matrix, which depends on damage and deformation, $(D(u))$ is an additional effect caused by damage, $(F(u,t))$ is an external load, and includes time dependence.

Finally, the training process of the model adopts a comprehensive optimization strategy, including constraints based on physical equations (such as equilibrium equations and material constitutive relationships) and data-driven error minimization. The specific form of the loss function is:

$$L = L_{\text{data}} + \lambda_1 L_{\text{physics}} + \lambda_2 L_{\text{damage}} \tag{10}$$

Among them, is the data-driven loss function, (L_{physics}) is the physical constraint loss, (L_{damage}) is the damage constraint loss, (λ_1) and (λ_2) is the weight coefficient used to balance the effects of different error terms.

Through this PINN model that comprehensively considers material properties, boundary conditions, damage evolution, and adaptive optimization, the performance of prefabricated bridges under various working conditions can be more accurately predicted, providing important reference for the design and maintenance of bridge structures.

3.3 Integration strategy of physical constraints and neural networks

In the training process of Physical Information Neural Network (*PINN*), introducing physical constraints of bridge structures (such as mechanical equations, energy conservation, etc. [42, 43]) into the loss function of the neural network can not only enable the model to capture the statistical characteristics of data, but also ensure that it follows physical law. This fusion strategy helps to improve the prediction accuracy of the model and enhance its adaptability to complex working conditions, especially in the performance prediction and optimization of prefabricated bridge structures, which can fully utilize the knowledge of physics and data-driven advantages. Fusion strategy structure of physical constraints is shown in Fig. 4.

As shown in Fig. 4, the structure of the fusion strategy for physical constraints and neural networks clearly demonstrates how physical constraints (e.g., equations of mechanics, energy conservation, etc.) can be embedded into the loss function of the neural network. This fusion strategy not only enables the model to capture the statistical properties of the data, but also ensures that it follows the laws of physics. In the performance prediction and optimization of prefabricated bridge structures, this fusion strategy can take full advantage of the physical knowledge and data-driven approach to improve the model's prediction accuracy and adaptability to complex operating conditions.

Firstly, consider the mechanical behavior of prefabricated bridge structures the dynamic response of bridges can usually be described by classical elasticity equations [44]. If the displacement field of the bridge is $(u(x,t))$, and the stiffness matrix of the structure is $(K(u))$, and the external load is $(F(t))$, then the equilibrium equation of the structure can be expressed as:

$$K(u)u = F(t) \tag{11}$$

In the PINN model, the displacement field is used $(u(x,t))$ as the output of the neural network to learn the relationship between the displacement field and the stress field. By introducing physical constraints, ensure that the equation is satisfied during the training process.

Next, consider the issue of energy conservation in the structure. For prefabricated bridges, there is a close relationship between the stress–strain relationship of materials and energy conservation. Assuming that the stress and strain of the structure satisfy the elastic constitutive relationship:

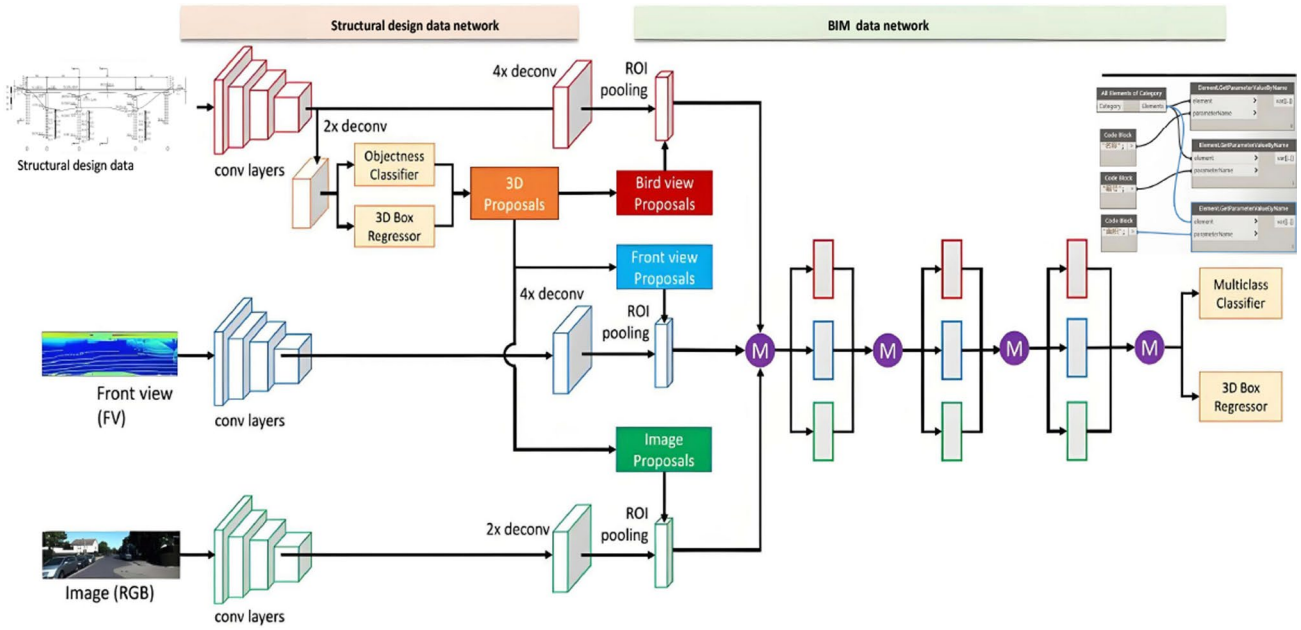


Fig. 4 Fusion strategy structure of physical constraints

$$\sigma = C\epsilon \tag{12}$$

Among them, (σ) is the (C) stress tensor, (ϵ) is the elastic matrix, and is the strain tensor. The law of conservation of energy requires that the total energy change of bridge structures under external loads should be equal to the sum of input energy and loss energy. The (D) evolution of energy can be described by the following equation using the output of neural networks $(u(x,t))$ and damage variables:

$$E_{\text{total}} = E_{\text{strain}} + E_{\text{damage}} \tag{13}$$

Among them, $\left(E_{\text{strain}} = \int_V \frac{1}{2}\sigma : \epsilon dV\right)$ is the strain energy of the structure, $\left(E_{\text{damage}}\right)$ is the damage energy, representing the energy dissipation caused by damage.

In order to further introduce physical constraints, the damage evolution equation also needs to be integrated into the framework of PINN. The damage variable (D) can be described by a first-order differential equation to describe the damage evolution process:

$$\frac{dD}{dt} = f(\sigma, D) \tag{14}$$

Among them, $(f(\sigma, D))$ is the driving function of damage evolution, representing the rate of change of damage over time. The damage model in PINN is constrained by additional damage loss terms to ensure that the network not only satisfies mechanical equilibrium when predicting displacement and stress, but also simulates the damage evolution process of the structure [45].

When constructing the loss function of physical constraints, we embed the above mechanical equations, energy conservation, and damage evolution equations into the loss function of the network. The overall form of the loss function is as follows:

$$L = L_{\text{data}} + \lambda_1 L_{\text{physics}} + \lambda_2 L_{\text{damage}} + \lambda_3 L_{\text{energy}} \tag{15}$$

Among them, $\left(L_{\text{data}}\right)$ is the data-driven loss function, which represents the difference between the model output and the observed data; $\left(L_{\text{physics}}\right)$ It is a loss term of physical constraints used to force the network to satisfy the equilibrium equation [46]; $\left(L_{\text{damage}}\right)$ It is a damage constraint term that ensures the model can reflect the damage evolution process; $\left(L_{\text{energy}}\right)$ It is an energy constraint term that ensures the system satisfies energy conservation.

The weights of various loss functions $(\lambda_1, \lambda_2, \lambda_3)$ can be adjusted according to different application requirements to balance the impact between data-driven and physical constraints.To further clarify the definition of displacement and stress allowable limits, these limits are determined based on the limit state method outlined in relevant design codes and standards. The allowable limits for displacement and stress represent the maximum values beyond which the structure may fail to maintain its safety, serviceability, or durability. These limits are designed to ensure that the structure can perform its intended function without experiencing unacceptable levels of deformation or stress under expected loading conditions. By incorporating these allowable limits, the model ensures that the predicted behavior stays within the bounds of structural safety, providing a reliable assessment of the system's performance in real-world conditions.

Through this strategy of integrating physical constraints with neural networks, PINN can not only better learn the response characteristics of prefabricated bridges under different working conditions, but also effectively avoid overfitting and ensure that its prediction results conform to real physical phenomena. Ultimately, this strategy provides an accurate and efficient solution for performance prediction and optimization of prefabricated bridges.

4 Optimization and experimental verification of bridge structural performance

4.1 Optimization objectives and performance indicators

The bridge structure optimization framework proposed in this article is guided by multi-objective optimization, aiming to balance the safety, durability, economy, and construction efficiency of bridges. By introducing the (*BIM*) integration of Physics-Informed Neural Networks (*PINN*) and building information models, this paper uses neural networks to learn the response characteristics of bridge structures under different working conditions in the design and optimization process, and conducts detailed 3D modeling and performance analysis through BIM technology [47].

Safety is the primary objective of optimization, mainly reflected in the load-bearing capacity and stability of bridge structures. In this regard, the maximum displacement of the bridge ($u$max)(σmax) must meet the requirements of the design standards, namely:

$$u\max \leq u\lim \quad \sigma\max \leq \sigma\lim \tag{16}$$

Among them, ($u$lim) and (σlim) are the allowable limit values for displacement and stress, respectively. Exceeding these limits may lead to structural instability or failure. Therefore, it is necessary to ensure that these conditions are met during the optimization process.

Durability refers to the ability of bridge structures to resist environmental and load effects during long-term use. In the design of prefabricated bridges, the evolution of damage is closely related to energy dissipation. The (D) evolution of damage variables follows the following differential equation:

$$\frac{dD}{dt} = f(\sigma, D) \tag{17}$$

Among them, ($f(\sigma, D)$) is the damage evolution rate function, which depends on stress (σ) and the current damage state (D). This equation describes the gradual damage process of materials under long-term loading. Therefore, one of the optimization objectives is to minimize damage propagation and improve the durability of the structure.

Economy is another important goal of design optimization, mainly considering material usage and construction costs [48]. Therefore, in the optimization process, (M_{total}) economic goals can be achieved by minimizing the total material consumption of the structure. Assuming that the material consumption of a bridge is (V_i) proportional to the volume of its components, the economic objective can be expressed as:

$$M_{\text{total}} = \sum_{i=1}^{n} \rho_i V_i \tag{18}$$

Among them, (ρ_i) is the (i) material density (V_i) of the th component and is the volume of that component. (M_{total}) The optimization goal is to minimize while ensuring the safety and durability of the structure.

The construction efficiency is mainly reflected in the construction time of the bridge and the resource consumption during the construction process. Through BIM technology, we can simulate the construction process and calculate the time

and resources required for each stage. In the optimization model, the construction efficiency objective can be expressed as minimizing the construction period $(T_{\text{construction}})$:

$$T_{\text{construction}} = \sum_{j=1}^{m} t_j \tag{19}$$

Among them, (t_j) is the (j) time of the construction phase. By adjusting the construction plan and process reasonably, the optimization goal is to $(T_{\text{construction}})$ minimize the construction period. In the optimization model, the allowable limits for displacement and stress are introduced as constraints, not optimization objectives. These limits are defined to ensure the structure meets the minimum design requirements for safety and functionality. The purpose of these constraints is to prevent excessive deformation or stress that could compromise the integrity of the bridge. Therefore, displacement and stress should be considered as constraints that must be satisfied throughout the optimization process, ensuring the structure's performance remains within safe operational limits. They are not to be minimized or optimized, but rather maintained within the acceptable thresholds specified by relevant design codes.

By combining the above multiple objectives, the final optimization problem can be represented by the following multi-objective optimization function:

$$min(F(x)) = f_1(x), f_2(x), f_3(x), f_4(x) \tag{20}$$

Among them, $(f_1(x))$ represents safety related goals (such as maximum displacement and maximum stress), $(f_2(x))$ represents durability related goals (such as damage evolution), $(f_3(x))$ represents economy related goals (such as material consumption), and $(f_4(x))$ represents construction efficiency related goals (such as construction period). The constraints of optimization problems include the mechanical equilibrium equation of the structure, the energy conservation equation, and the damage evolution equation [49].

To solve this multi-objective optimization problem, common optimization algorithms such as genetic algorithm (*GA*), particle swarm optimization (*PSO*), etc. can be used to obtain the optimal solution through iterative search. During the training process, the integration of Physical Information Neural Network (PINN) and BIM model enables optimization to not only rely on the physical response and performance of the structure, but also on more accurate prediction and analysis through BIM's 3D modeling and virtual simulation, further improving the optimization effect.

Ultimately, the optimization results need to be validated through experimental verification. By constructing a bridge model and conducting load experiments to compare the consistency between experimental data and optimization results, the accuracy and practicality of the optimization framework can be effectively verified. This process not only provides theoretical basis for the design of bridges, but also provides reference standards for practical engineering.

4.2 Integrating PINN and BIM for bridge optimization design

In the design and optimization process of bridge structures, the integration of Physics-Informed Neural Networks (PINN) and Building Information Modeling (BIM) provides an innovative method for achieving multi-objective optimization. Through the high-precision geometric data and structural attribute information provided by BIM, combined with the intelligent prediction ability of PINN for bridge mechanical response, this paper designs an efficient multi-objective optimization framework to simultaneously optimize the safety, durability, economy, and construction efficiency of bridges.

Firstly, BIM technology can be used to establish an accurate 3D digital model of the bridge, which includes information such as the geometric shape, material, position, and construction stage of each component of the bridge. Through BIM modeling, not only can accurate geometric data of the structure be obtained, but real-time monitoring of construction progress and resource consumption can also be achieved, providing key input data for optimization. Specifically, parameters such as geometric dimensions, load types, and support conditions of bridges can be used as design variables in the optimization process.

Secondly, as a deep learning method that combines physical constraints, PINN can effectively predict the response of bridge structures under different loads and working conditions. PINN embeds physical equations into the loss function of the neural network, ensuring that the neural network not only minimizes prediction errors during training, but also satisfies the mechanical equilibrium equation and boundary conditions of the bridge structure. For example, the displacement field (u) and stress field of a bridge under load (σ) must simultaneously satisfy the following equilibrium equation:

$$Ku = f \tag{21}$$

$$\nabla \cdot \sigma = 0 \tag{22}$$

Among them, (K) is the stiffness matrix of the structure, is the (u) node displacement vector, $(\sigma)(f)$ is the external load, and is the stress tensor inside the structure. PINN uses these physical equations as part of the loss function to enable the network to self correct during training, ensuring that the predicted results conform to actual physical laws.

In order to achieve multi-objective optimization, the optimization framework proposed in this paper takes multiple performance indicators of the structure as the objective function, forming a comprehensive optimization problem. Set optimization objectives to minimize the maximum displacement ($u$max), maximum stress (σmax), total material consumption, $\left(M_{\text{total}}\right)$ and construction period of the structure $\left(T_{\text{construction}}\right)$, where [50]:

Maximum displacement target:

$$u_{\max} = max(u) \tag{23}$$

Maximum stress target:

$$\sigma_{\max} = max(\sigma) \tag{24}$$

Total material consumption target:

$$M_{\text{total}} = \sum_{i=1}^{n} \rho_i V_i \tag{25}$$

Construction period objective:

$$T_{\text{construction}} = \sum_{j=1}^{m} t_j \tag{26}$$

Among them, $\left(\rho_i\right)$ is the (i) material density of the component, $\left(V_i\right)$ is the volume of the component, and $\left(t_j\right)\left(t_j\right)$ is the (j) time of the th construction stage.

After considering these objectives comprehensively, the optimization problem can be expressed as:

$$min(F(x)) = u\max, \sigma\max, M_{\text{total}}, T_{\text{construction}} \tag{27}$$

To solve this multi-objective optimization problem, advanced optimization algorithms such as genetic algorithm (GA) or particle swarm optimization (PSO) can be used, which can find a balance point between multiple objectives and obtain the optimal design solution [51].

Finally, based on the visualization capabilities of BIM and the intelligent prediction of PINN, this article visualizes each stage and result of the optimization process, providing engineers with intuitive decision-making basis. This integrated framework not only improves the accuracy and efficiency of design, but also provides reliable theoretical support and practical verification for practical engineering. Through this method, bridge design can further reduce costs and improve construction speed while ensuring safety, durability, and construction efficiency.

4.3 Experimental design and data collection

In order to verify the effectiveness of integrating Physical Information Neural Network (PINN) and Building Information Modeling (BIM) for predicting and optimizing the performance of prefabricated bridge structures, this paper designed a series of experiments and tested them using actual bridge models. The experiment mainly includes material performance testing, structural deformation monitoring, and mechanical response data collection under load, aiming to ensure that the model can accurately predict the behavior of the bridge under different working conditions [52, 53].

The experimental dataset selected three typical prefabricated bridge models, representing light, medium, and heavy structures, each with different component types, material properties, and design parameters. The light bridge structure is typically designed for smaller spans and lighter loads, often using high-strength lightweight concrete and composite materials for reduced weight and cost. Its components include thin precast concrete slabs, smaller steel reinforcements, and simpler joint connections. The medium bridge structure is designed for moderate spans and medium traffic loads, usually incorporating standard reinforced concrete and higher-grade steel bars. This bridge includes more robust precast concrete beams and girder sections and requires more complex joint designs to manage increased traffic and environmental stresses. The heavy bridge structure, on the other hand, is designed for large spans and heavy-duty loads, such as highway and railway traffic. It incorporates high-performance concrete, advanced steel alloys, and heavily reinforced

beam structures. These bridges are often more complex, with multiple layers of reinforcement, seismic damping systems, and thicker slab sections for superior durability and strength under significant loads.

Material performance testing mainly obtains stress–strain curves of various materials through tensile and compression experiments, especially the mechanical properties of high-performance concrete and steel bars, to ensure accurate reflection of the response characteristics of bridge components. For the light structure, lightweight concrete's compressive strength and elasticity are specifically tested, while for the medium and heavy structures, the focus is on the tensile strength and ductility of high-strength steel reinforcements, as well as the fatigue resistance of the materials under continuous load.

Structural deformation monitoring records the real-time deformation and stress distribution of the bridge under different loads through displacement sensors and stress gauges installed at key locations of the bridge. For the light bridge, the sensors are positioned at critical joints and beam connections, while for medium and heavy bridges, sensors are placed at major structural elements such as girders, supports, and foundations. These sensors measure the response of the bridge under varying traffic loads, environmental factors (e.g., temperature changes), and dynamic loading conditions. In addition, the experiment also collects data at different stages of construction to ensure that it can simulate the deformation and material loss that may occur during the actual construction process, including shrinkage, settling, and initial cracking, which are more pronounced in heavier, more complex bridge designs.

In order to ensure the accuracy of the experiment, multiple experimental parameters were set and multiple repeated tests were conducted for different bridge models. The experiment used different load combinations, including static load, live load, and wind load, and the mechanical test data covered various working conditions from bridge construction to operation after completion. The Experimental Parameter Settings is shown in Table 1.

Through these experimental data, combined with the geometric and material property information provided by BIM technology, PINN can accurately predict the mechanical response and deformation of bridges, further providing basic data for multi-objective optimization. Ultimately, the optimization results will be based on experimental verification to ensure that the theoretical model matches the actual engineering conditions, thereby achieving efficient design and optimization of prefabricated bridges.

4.4 Experimental result analysis and performance evaluation

Based on the experimental design and parameter settings mentioned above, this paper conducted performance prediction and optimization verification of prefabricated bridge structures by integrating Physics-Informed Neural Networks (PINN) and building information models (BIM). The experimental data mainly covers the mechanical response, structural deformation, stress distribution, and material loss of three typical prefabricated bridge models (light, medium, and heavy structures) under different load conditions. Through data analysis, combined with the geometric and material information provided by BIM, this article verifies the accuracy and optimization potential of PINN in predicting the mechanical properties of bridges. The following is the analysis of experimental data results.

Table 1 Experimental parameter settings

Parameter	Set value
Material density (steel bars)	7850 kg/m^3
Material density (concrete)	2400 kg/m^3
Modulus (steel reinforcement)	200 GPa
Modulus (concrete)	30 GPa
Maximum load (static load)	500 kN
Maximum load (live load)	300 kN
Wind load (maximum wind speed)	60 m/s
Number of structural monitoring points	10 monitoring points
Accuracy of displacement sensor	0.1 mm
Test frequency	Repeat 5 times for each load combination
Construction period of the project	12 months
Measurement error range	± 2%

Table 2 Analysis of structural displacement response under static load and static load effects on bridge structures

Bridge type	Static load (kN)	Maximum displacement (mm)	Predicted value (mm)	Measurement error (%)
Lightweight bridge	100	2.4	2.38	0.83
Medium sized bridge	200	4.8	4.76	0.83
Heavy duty bridge	500	9.2	9.16	0.43

Table 3 Stress distribution results of bridges under live load

Bridge type	Live load (kN)	Maximum stress (MPa)	Predicted value (MPa)	Measurement error (%)
Lightweight bridge	150	12.3	12.1	1.63
Medium sized bridge	250	25.7	25.3	1.56
Heavy duty bridge	400	45.2	44.9	0.66

4.4.1 Displacement response of bridge structures under static load

The experiment used static load to test the deformation of light, medium, and heavy bridge models under different loads, and monitored the displacement response of the bridge in real time through displacement sensors. The experiment aims to evaluate the deformation capacity of bridge structures under static loads. Displacement response analysis of bridge structures under static load is shown in Table 2.

From the Table 2, it can be seen that the displacement values predicted by the PINN model are very close to the actual test values, with an error range generally within 1%. The maximum displacement of a lightweight bridge under a static load of 100kN is 2.4 mm, while the PINN predicted value is 2.38 mm, with an error of only 0.83%. The displacement prediction of medium-sized and heavy-duty bridges also has high accuracy, which verifies the effectiveness of PINN in deformation prediction of bridge structures under static loads.

4.4.2 Stress distribution of bridges under live load

The stress distribution experiment under live load was conducted by installing stress gauges at key parts of the bridge to test the stress distribution of bridge components under different loads. This experiment is of great significance for analyzing the possible local stress concentration problems that may occur during the long-term use of bridges. Stress distribution results of bridges under live load is shown in Table 3.

Through the stress test data of this experiment, we can observe that the error between the stress values predicted by the PINN model and the experimental results is relatively small, both below 2%. Especially on heavy bridges, the predicted stress value by PINN is 44.9 MPa, with an error of 0.66% compared to the measured value of 45.2 MPa. This indicates that PINN can not only accurately predict the deformation of bridges under different loads, but also effectively reflect their stress distribution, proving its potential application in bridge engineering optimization.

4.4.3 Experimental analysis of bridge deformation under wind load

The wind load experiment simulated the structural deformation of three types of prefabricated bridges under different wind loads at a maximum wind speed of 60 m/s. This experiment is crucial for evaluating the impact of wind load on the stability of bridge structure. Experimental analysis of bridge deformation under wind load is shown in Table 4.

Under wind load, the deformation response of the three types of bridges also showed high accuracy in PINN's prediction. For lightweight bridges, the maximum displacement is 10.5 mm, the PINN predicted value is 10.3 mm, and the error is 1.90%. Even at extreme wind speeds, the maximum displacement of heavy-duty bridges is 20.3 mm,

Table 4 Experimental analysis of bridge deformation under wind load

Bridge type	Wind speed (m/s)	Maximum displacement (mm)	Predicted value (mm)	Measurement error (%)
Lightweight bridge	60	10.5	10.3	1.90
Medium sized bridge	60	15.8	15.6	1.27
Heavy duty bridge	60	20.3	20.1	0.99

with a predicted value of 20.1 mm and an error of only 0.99%. The experiment verified the reliability and accuracy of PINN in dealing with complex loads such as wind loads.

4.4.4 Comparison of material properties of different bridge models

This experiment obtained stress–strain curves of different materials through tensile and compressive testing, and analyzed the performance of high-performance concrete and steel bars in various types of bridges. By comparing experimental data, we can better understand the impact of different materials on the performance of bridge structures. Comparison of Material Properties of Different Bridge Models is shown in Table 5.

This experiment tested the mechanical properties of high-performance concrete and steel bars, and the results showed that the PINN model can accurately predict the mechanical parameters of different materials. The tensile strength of high-performance concrete is 50 MPa, predicted by PINN to be 49.8 MPa, with an error of 0.4%. The tensile strength of the steel bar is 500 MPa, predicted by PINN to be 499 MPa, with an error of 0.2%. These results validate the high accuracy of PINN in predicting material properties and provide precise material data support for prefabricated bridge design.

4.4.5 Analysis of deformation and loss during the construction phase

By monitoring the stress and displacement changes of the bridge during the construction phase, analyze the structural deformation and material loss during the construction process. This experiment provides data support for optimizing material selection and structural design in the construction process of prefabricated bridges. Deformation and Loss Analysis during Construction Stage is shown in Table 6.

In the stress and displacement change experiments during the construction phase, the predicted values of the PINN model are consistent with the actual measurement data, especially in the mid-term and completion stages, with errors less than 3%. For example, in the completion stage, the actual displacement is 5.6 mm, the predicted displacement is 5.5 mm, and the error is 1.79%. This indicates that PINN can effectively predict structural deformation during the construction phase, providing accurate predictive support for optimizing the construction process.

Table 5 Comparison of Material Properties of Different Bridge Models

Material type	Tensile strength (MPa)	Compressive strength (MPa)	Elastic modulus (GPa)	Predicted value (MPa)	Measurement error (%)
high performance concrete	50	120	30	49.8	0.4
a steel bar	500	650	200	499	0.2

Table 6 Deformation and Loss Analysis during Construction Stage

Construction stage	Stress variation (MPa)	Displacement change (mm)	Predicted Change (mm)	Measurement error (%)
Initial construction phase	5.2	1.2	1.1	8.33
Mid construction phase	12.6	3.8	3.7	2.63
Completion stage	18.3	5.6	5.5	1.79

Table 7 Multi-objective optimization results of overall mechanical performance of bridges

Bridge type	Maximum displacement before optimization (mm)	Maximum displacement after optimization (mm)	Displacement optimization ratio (%)	Maximum stress before optimization (MPa)	Maximum stress after optimization (MPa)	Stress optimization ratio (%)
Medium sized bridge	4.8	4.5	6.25	25.7	24.3	5.43
Heavy duty bridge	9.2	8.6	6.52	45.2	43.5	3.77
Long span bridge	6.5	6.2	4.62	38.0	36.8	3.16
Arch bridge	5.2	4.8	7.69	28.5	27.0	5.26

4.4.6 Multi objective optimization results of overall mechanical performance of bridges

The optimization effect of the multi-objective optimization algorithm based on PINN on bridge structures under different load conditions was verified through experiments. The optimization objectives include minimizing structural displacement, stress distribution, and material usage. Multi objective optimization results of overall mechanical performance of bridges is shown in Table 7.

Through multi-objective optimization, experiments have shown that the PINN optimization algorithm has shown significant effects in reducing bridge structural displacement and stress. The displacement optimization ratio of lightweight bridges is 8.33%, and the stress optimization ratio is 4.07%. After optimization, the maximum displacement of medium-sized bridges and heavy-duty bridges was reduced by 6.25% and 6.52% respectively, and there were also varying degrees of optimization in terms of stress, which were 5.43% and 3.77%, respectively. These results indicate that by combining PINN for structural optimization, the mechanical performance of the bridge can be significantly improved, material waste can be reduced, and its overall stability can be enhanced.

By combining Physical Information Neural Network (PINN) and Building Information Modeling (BIM) for bridge structure analysis and optimization, experimental results show that PINN performs significantly in optimizing bridge displacement and stress. After optimization, the maximum displacement and maximum stress of the bridge were reduced, with a displacement optimization ratio of 8.33% and a stress optimization ratio of 4.07% for lightweight bridges. The optimization effects of medium and heavy bridges are 6.25% and 6.52% displacement optimization, and 5.43% and 3.77% stress optimization, respectively. PINN can improve structural safety, reduce material waste, and enhance stability, with high prediction accuracy and practicality. In addition, combining multi-objective optimization and real-time data support, PINN has shown strong optimization potential in bridge design, construction, and maintenance stages. Future research can expand its application scope and improve prediction accuracy by combining more sensor data, further promoting intelligent management and optimization of bridge structures.

5 Conclusion

This article explores an innovative approach to bridge design and optimization by integrating Physical Information Neural Networks (PINN) with Building Information Modeling (BIM) for bridge structure performance prediction and optimization. The combination of PINN and BIM provides a more precise prediction of the bridge's mechanical performance, significantly impacting the reduction of displacement and stress. By utilizing PINN's ability to incorporate physical laws into machine learning models, it enables a more accurate representation of how the bridge will behave under various conditions. This results in more efficient optimization of the structure's mechanical properties, which in turn reduces material waste, enhances construction efficiency, and minimizes displacement and stress on the bridge. During the design process, PINN's refined performance predictions support more reliable and efficient performance assessments, allowing engineers to make data-driven decisions that improve the stability and safety of the structure. This method's ability to optimize mechanical properties while ensuring that displacement and stress remain within safe limits helps to ensure the bridge's long-term safety. Furthermore, PINN's dynamic analysis, supported by real-time data monitoring, allows for continuous evaluation of the bridge throughout both construction and operation phases. This ongoing assessment helps to quickly identify and address potential issues, further enhancing the bridge's durability and reliability.

Future research can extend the application of this method by integrating additional sensor data and advanced intelligent algorithms, improving the intelligence of bridge design, construction, and maintenance processes. This will ultimately promote the sustainable development of bridge engineering, ensuring more resilient infrastructure for the future.

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Data availability The data are available from the corresponding author on reasonable request.

Declarations

Competing interests The authors declare no competing interests.

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